FarmTwin

Agri Digital-Twin & Precision-Operations Platform

Simulate every field. Irrigate by need, not by habit.

Live pilot: 15-acre farm, Eruthempathy, Chittur (Palakkad) | KSUM / AgriNext

The Problem

Kerala's rain-shadow belt: water arrived, scheduling did not.

- Chittur (Palakkad) gets ~1,000-1,250 mm rain vs the state's ~3,000 mm.
- The 2025 Moolathara canal extension now serves ~3,575 ha with drip/lift water.
- But farmers still irrigate by calendar and habit, not by crop-and-soil need:
  - Over-irrigation wastes the scarce canal water just secured.
  - Under-irrigation causes crop stress and yield loss.
- No plot-level, day-by-day decision tool exists for this microclimate.
- FPOs have no shared layer to budget water or plan harvests across 650+ ha.

Quantified Pain

Why the bottleneck is now optimization, not availability.

30-50%

of applied water wasted by flood/over-irrigation vs precise scheduling

10-25%

of potential yield lost to mistimed irrigation & nutrient stress

~30%

of manual irrigation labour reducible with scheduled precision

The Solution

A farm digital twin built on a physics + agronomy engine.

- Discretize a field into a computational mesh of zones - like a finite-element mesh.
- A time-stepped water-balance + crop-growth solver advances each zone, day by day.
- Ingests soil type, crop, emitter layout, weather/ET and low-cost soil-moisture data.
- Answers per zone, per day: irrigate or not, how many litres, and why.
- Optimizes the irrigation/fertigation schedule to minimize water & cost at target yield.
- Run what-if scenarios (flood vs drip vs optimized, heat wave) before acting in the field.

Why It's Defensible

20 years of CAE / simulation, re-pointed at agriculture.

CAE / simulation skill	FarmTwin application
Meshing algorithms	Discretize the farm into adaptive irrigation/soil zones
FE / CFD field solvers	Water-balance + moisture & nutrient transport across the mesh
Assembly-line simulation	Time-stepped scheduling under a daily water budget
DOE & optimization	What-if scenarios + optimal irrigation/fertigation schedules
Validation vs test data	Calibrate the twin against sensors & observed yield

The moat is the calibrated simulation engine - not another IoT dashboard.

Two Products, One Shared Engine

A calibration feedback loop that compounds over time.

FarmTwin Studio (pre-install)	FarmTwin Runtime (operations)
Used once, before installation	Runs continuously, after installation
Survey -> simulate -> optimize	Sense -> decide -> actuate
Recommends top 2-3 designs, BoM	Live control, twin state, alerts, yield
Designer / agronomist / dealer	Farmer / FPO operator (autonomous)

Shared FarmTwin Engine (krishiflow): Runtime re-estimates parameters and feeds Studio - designs keep improving.

Product & MVP

A working prototype already demonstrates the core value.

- Web app: a farm is a mesh of zones; each zone is modelled and solved daily.
- Scenario comparison - flood vs uniform drip vs twin-optimized.
- Outputs: water saved, stress-days avoided, projected yield.
- Hardware-light: cheap/existing soil & weather sensors + free weather/ET data.
- MVP runs in any browser (mvp/index.html) - no build step.

Market & Beachhead

One FPO sale reaches hundreds of farmers.

- Primary pilot: founder's 15-acre farm + 8-12 neighbouring vegetable/mango farmers.
- Beachhead buyer: FPOs / FPCs in the Chittur rain-shadow cluster (1,500+ farmers, 650 ha).
    - Routed through the World Bank KERA / AgriNext program.
- Scale buyer: Krishi Bhavans & Dept. of Agriculture in other rain-shadow districts.
- Phase 2: lenders, insurers and input suppliers buying the data/risk layer.

Why Now

Three tailwinds line up in 2025-2026.

- Canal water arrived in 2025 - scheduling is suddenly the binding constraint.
- World Bank KERA / AgriNext (2026) is funding & distributing exactly this category to FPOs.
- Cheap soil/weather sensing + free ET data make a software-led, hardware-light product viable.

Pilot & Traction Plan

Milestone-disbursed, verifiable success criteria.

- M1 (month 2): sensors deployed on pilot farm; twin ingesting live data.
- M2 (month 4): solver calibrated to a defined soil-moisture error band.
- M3 (month 6): scenario engine shows >=30% water saving; 1 signed FPO pilot MoU.
- Target: >=30% simulated water saving vs flood baseline at equal yield.

Funding Roadmap

Sequence non-dilutive money first; prove traction, then scale.

- 1. Idea Grant - up to Rs.3L (MVP + pilot plan).
- 2. AgriNext / KERA pilot - up to Rs.25L (FPO match).
- 3. Productisation Grant - up to Rs.7L (MVP -> product).
- 4. Seed Fund / Seed Loan - up to Rs.15L (soft loan vs orders).
- 5. Scale-up Seed Fund - up to Rs.25L (repeatable paid deployments).
- 6. KERA Productive Alliance - up to Rs.2cr (FPC alliance, 60% matching).

Use of Funds & The Ask

KSUM Idea Grant: Rs.3,00,000

Item	Amount (Rs.)
Field instrumentation (sensors, gateway)	90,000
Cloud + dev tooling (12 months)	60,000
Solver/MVP hardening (contractor time)	90,000
Calibration & field validation	45,000
IP / legal (patent search, incorporation)	15,000

Ask: Rs.3 lakh Idea Grant + KSUM incubation + AgriNext deployment introductions.

FarmTwin

I have spent 20 years meshing and simulating physical systems.

A farm is just another domain to mesh and solve - and I own the test rig.